

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Justification

MILU's 8-domain, 41-subject taxonomy was designed around Indian competitive exam content [Q22, Q42] and meaningfully overlaps with the deployment's priority subjects: Polity (Law & Governance), History (Social Sciences/Arts & Humanities), Economy (Business Studies), and GK. Dataset analysis empirically confirms strong representation of UPSC-relevant Polity, History, Economy, Hindi grammar, and even UPSC-style Ethics scenarios. However, MILU's macro-domain taxonomy does not separately enumerate Mathematics/Quantitative Aptitude, Logical Reasoning, or Current Affairs as named categories — three of the deployment's HIGH-priority subject areas for SSC/banking and UPSC CSAT. Dataset analysis partially mitigates this concern (reasoning items appear under Science domain, arithmetic under Business Studies), but pooling across MILU's 40+ exam types [Q27] means central-exam vs. state-PSC items are not separable, and the sample also reveals significant construct-irrelevant content (specialist Engineering/FORTRAN, South India-specific items) that is irrelevant to UPSC/SSC/banking syllabi.

Strengths

- Indian-exam-grounded taxonomy with explicit cultural subjects (local history, arts, festivals, laws) [Q12, Q42]
- Empirical coverage of Polity, History, Economy, Hindi grammar, and UPSC-style ethics confirmed in the Hindi validation split [DATASET-D1, D2, D9, D56]

ID	Evidence
D1	_____ एक अदालत का आदेश है जो एक अधीनस्थ अदालत या सार्वजनिक प्राधिकरण को एक आधिकारिक कर्तव्य को सही ढंग से नभाने का निर्देश देता है । — Mandamus writ definition question — Indian constitutional law
D2	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? — Constitutional amendment adding 'armed rebellion' — Indian Polity
D9	_____ ने दिल्ली में कुतुब मीनार का निर्माण पूरा किया । — Who completed Qutub Minar — Medieval Indian history
D56	एक वरिष्ठ सविलि अधिकारी के रूप में, आपको नज़ी रूप से संचालित वृद्धाश्रम के वार्षिक समारोह में अतिथि बनने के लिए आमंत्रित किया गया है... आदवासी लोग जर्जर झोपड़ियों में रह रहे थे — Ethics/governance scenario for senior civil officer — UPSC GS Paper IV Ethics

- Reasoning and quantitative content present in actual data despite not being named macro-domains [DATASET-D13, D15, D17, D18]

ID	Evidence
D13	रचि ने परीक्षा में नमिनलखित श्रृंखला दी है । TAB, TTZBB, TTBBB, TTTYBBB, TTTCBBB, _____. — Letter series completion — Logical Reasoning
D15	सूरज ने अजय से एक नश्चिती दर पर साधारण ब्याज पर दो वर्षों के लिए एक राशुधर ली और उसे वनिय को उसी दर पर वार्षिक चक्रवृद्धि ब्याज पर दो वर्षों के लिए उधार दिया । — Simple vs. compound interest calculation — Quantitative Aptitude/Mathematics
D17	आठ लोग दो समानांतर पंक्तियों में बैठे हैं... A के ठीक विपरीत कौन बैठा है? — Seating arrangement reasoning problem
D18	कथन: I. कुछ पेन कप हैं । II. सभी कप प्लेट हैं । निष्कर्ष: I. सभी पेन प्लेट हैं । — Syllogism — formal logical reasoning

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories

IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories per elicitation Q1: GK, Current Affairs, History, Mathematics/Reasoning, Polity, Hindi proficiency. MILU's taxonomy [Q42] explicitly covers History/Polity/GK domains; Mathematics/Reasoning and Current Affairs are not named macro-domains but appear empirically in data [DATASET-D13-D18, D19-D22].

ID	Evidence
WEB-9	UPSC Current Affairs ranges 0–27 questions per year — high weight, currency-sensitive
WEB-10	SSC CGL Tier 1 covers General Intelligence/Reasoning, General Awareness, Quantitative Aptitude, English Comprehension
WEB-15	UPSC CSAT Paper 2 covers Quantitative Aptitude and Logical Reasoning

IO-2: Mathematics/Quantitative Aptitude, Logical Reasoning, and Current Affairs are not enumerated in the 8-domain taxonomy [Q42]; this is a documented partial gap, mitigated by empirical presence of reasoning/arithmetic items in the Hindi split [DATASET-D13, D15, D16, D17, D18].

ID	Evidence
D13	रक्षा ने परीक्षा में नमिनलखिति श्रृंखला दी है । TAB, TTZBB, TTBBB, TTTYBBB, TTTCBBB, _____. — Letter series completion — Logical Reasoning
D15	सुरज ने अजय से एक निश्चित दर पर साधारण ब्याज पर दो वर्षों के लिए एक राशिधार ली और उसे वनिय को उसी दर पर वार्षिक चक्रवृद्धि ब्याज पर दो वर्षों के लिए उधार दिया । — Simple vs. compound interest calculation — Quantitative Aptitude/Mathematics
D16	किसी कंपनी में सभी कर्मचारियों का औसत वेतन रु. 10500 है । सभी पुरुष कर्मचारियों का औसत वेतन रु. 15000 है । — Average salary mixture problem — Quantitative Aptitude
D17	आठ लोग दो समानांतर पंक्तियों में बैठे हैं... A के ठीक विपरीत कौन बैठा है? — Seating arrangement reasoning problem
D18	कथन: I. कुछ पेन कप हैं । II. सभी कप प्लेट हैं । निष्कर्ष: I. सभी पेन प्लेट हैं । — Syllogism — formal logical reasoning

IO-3: Sample contains significant Engineering & Technology specialist content (DC motors, FORTRAN, AM/FM modulation, rectifiers) outside UPSC/SSC/banking syllabi [DATASET-D35, D36, D37, D38]; this is construct-irrelevant variance for the deployment.

ID	Evidence
D35	जब एक डीसी सीरीज मोटर बना लोड के चलती है: मोटर की गति बहुत अधिक होती है — DC series motor behavior — Engineering, not UPSC/SSC priority
D36	हाफ वेव रेक्टिफायर का आउटपुट क्या होता है: पल्सेटिंग डीसी — Half-wave rectifier output — Electrical engineering technical content
D37	फॉरट्रान 77 के फिक्स्ड फॉर्मेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? — FORTRAN 77 fixed format — specialist programming, not UPSC/SSC priority
D38	540 kHz से 1650 kHz तक ट्यूनिंग करने वाले रिसीवर के लिए सबसे लोकप्रिय इंटरमीडिएट फ्रीक्वेंसी (kHz में) कौन सी है — Intermediate frequency for AM receivers — electronics engineering specialist content

IO-4: Pooling of 40+ exam types [Q27] across national and state levels [Q92] without published per-tier breakdown means central-exam items cannot be isolated; this harms content validity for a central-exam-only deployment.

Information Gaps

- Proportion of MILU Hindi items sourced from central exams (UPSC/SSC/banking) vs. state PSCs is not published
- Item-level breakdown of Current Affairs vs. timeless GK is unavailable

Requires Expert Verification

- Whether the empirical density of reasoning and quantitative items in the full Hindi test split is sufficient for SSC/banking evaluation

Remediation

Gap 1: Pooled 40+ exam types [Q27] and presence of specialist Engineering/South India content [DATASET-D35-D38, D31-D34] dilute central-exam relevance

Recommendation: Create a deployment-specific subset by filtering MILU items to Polity/Law&Governance, Social Sciences, Business Studies arithmetic, Science reasoning items, and Hindi grammar; explicitly drop Engineering & Technology specialist items not in UPSC/SSC/banking syllabi.